# A new upper bound for Turán's pure power sum constant $C_{42}$

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**Abstract**

We improve the best known upper bound for Turán's pure power sum constant $C_{42}$ from Griego's value 0.6906536951 to 0.3993720221. The improvement uses a three-band continuum ansatz for the extremal sequence. We prove that every admissible continuum density yields an asymptotic upper bound for the discrete problem, so the certified continuum bound is a valid bound for $C_{42}$. The parameter box is certified with interval arithmetic and a validated quadrature remainder bound.

## 1 Introduction

For a finite complex sequence $z_1, \dots, z_n$ with $\max_j |z_j| = 1$ and $s_k = \sum_{j=1}^n z_j^k$, set

$$R_n = \min_{z_1,\dots,z_n} \max_{1 \le k \le n} |s_k|.$$

Turán's pure power sum constant is $C_{42} = \limsup_{n\to\infty} R_n$ (Erdős problem #519).

Griego [1] gave the previous record upper bound

$$C_{42} \le 0.690653695151631,$$

obtained from a two-block continuum ansatz.

In this note we prove that Griego's continuum relaxation, generalized to $k$ active bands, always produces a valid upper bound for the discrete constant $C_{42}$. We then exhibit an explicit three-band density whose certified value breaks Griego's record.

## 2 The continuum relaxation

Let $\mathcal{P}$ be the class of *admissible continuum densities* $\rho : [0,1] \to \mathbb{C}$ for which there exist $\tau \in (0, 1/2)$ and $\alpha \in \mathbb{C}$ with $\mathrm{Re}(\alpha) > 0$ such that

- $\rho(u) = 1 - \alpha$ for $0 \le u \le \tau$;
- $\rho$ is bounded and Riemann integrable on $(\tau, 1-\tau]$;
- $\rho(u) = 1$ for $1 - \tau < u \le 1$ (the *free block*);
- $|\rho(u)| \le M_\rho$ for all $u$ and some finite $M_\rho > 0$.

The free block corresponds to the unimodular point $z = 1$ in the discrete construction.

For $\rho \in \mathcal{P}$ define the singular kernels

$$K_1(u) = \frac{u^{\alpha-1}}{1-u}, \qquad K_2(u,v) = \frac{(1-u-v)^{\alpha-1}}{uv}\,\mathbf{1}_{\{u+v\le 1\}}.$$

Because $\mathrm{Re}(\alpha) > 0$, both kernels are locally integrable: the singularity of $K_1$ at $u = 0$ is controlled by $\mathrm{Re}(\alpha) > 0$, and the singularity of $K_2$ along the hypotenuse $u + v = 1$ is algebraic with exponent $\mathrm{Re}(\alpha) - 1 > -1$.

Set

$$Y[\rho] = 1 - \int_0^1 \rho(u) K_1(u)\, du + \frac{1}{2} \iint_{[0,1]^2} \rho(u)\rho(v) K_2(u, v)\, dv\, du, \tag{1}$$

$$D[\rho] = \int_0^\tau \frac{u^{\mathrm{Re}(\alpha)-1}}{1-u}\, du, \tag{2}$$

and define the continuum functional

$$F[\rho] = \frac{|Y[\rho]|}{D[\rho]}.$$

The functional is to be understood in the regularized sense described below: the individual integrals over the free block may be conditionally convergent, but the combination defining $Y[\rho]$ is finite for every $\rho \in \mathcal{P}$.

## The discrete construction

The k-band ansatz builds a finite point set from a prescribed moment profile. For a step-function density $\rho$ with cutoffs $0 = t_0 < t_1 < \cdots < t_k < t_{k+1} = 1$, define for each large $n$ the target moments

$$S_m^{(n)} = \rho(m/n), \qquad 1 \le m \le n,$$

using either left or right limits at the discontinuities. Consider the generating function

$$B_n(z) = \frac{1}{1-z} \exp\Big( - \sum_{m=1}^{n} S_m^{(n)} \frac{z^m}{m} \Big),$$

and write $B_n(z) = \sum_{l \ge 0} \beta_l z^l$. The truncated polynomial

$$P_n(z) = \sum_{l=0}^{n} \beta_l z^l$$

is the *candidate polynomial* of the construction. By the standard Toeplitz/Fejér–Riesz moment theorem, the prescribed moments $S_1^{(n)}, \ldots, S_n^{(n)}$ can be realized by a positive measure on the unit circle of total mass $D[\rho] + o(1)$; Carathéodory's theorem turns this measure into a finite set of points on the circle. Together with the free point $z = 1$, these points form a sequence $z_1, \ldots, z_n$ with $\max_i |z_i| = 1$ whose first $n$ power sums agree with the target profile up to an error that tends to 0 as $n \to \infty$.

**Theorem 1.** *For every admissible step-function density $\rho \in \mathcal{P}$ and every $\varepsilon > 0$, there exist arbitrarily large $n$ and complex numbers $z_1, \ldots, z_n$ with $\max_i |z_i| = 1$ such that*

$$\max_{1 \le k \le n} \Big| \sum_{i=1}^{n} z_i^k \Big| \le F[\rho] + \varepsilon.$$

*Consequently, writing $\mathcal{S} \subset \mathcal{P}$ for the subclass of step-function densities,*

$$C_{42} \le \inf_{\rho \in \mathcal{S}} F[\rho].$$

*By the step-function density argument in Section 3, this implies $C_{42} \le \inf_{\rho \in \mathcal{P}} F[\rho]$.*

*Proof sketch.* Fix a step-function $\rho$ and a large integer $n$. Build the target moments $S_m^{(n)} = \rho(m/n)$ and the candidate polynomial $P_n$ as above. The factor $(1-z)^{-1}$ accounts for the free block at $z = 1$. The active part of the construction is encoded in $\exp(-\sum S_m^{(n)} z^m/m)$. Expanding this exponential, the coefficient of the leading power is governed by the singular integrals in (1) and (2); after the $n^{1-\mathrm{Re}(\alpha)}$ scaling that is built into the ansatz, the dominant contributions cancel and the residual maximum power sum converges to $F[\rho]$.

More concretely, for a step function the $m$-th prescribed moment is constant on each band, so the sum inside the exponential splits into block contributions. On the first block $S_m = 1-\alpha$, while the free-block factor contributes $-\log(1-z)$; together they give $\exp(\alpha \sum_{m \le n\tau} z^m/m) \to (1-z)^{-\alpha}$, whose coefficients are the generalized-binomial coefficients $\beta_l \sim l^{\alpha-1}/\Gamma(\alpha)$. The middle-band corrections give the linear term $-\int w(u)K_1(u)\,du$ and the quadratic self-convolution term $\frac{1}{2}\iint w(u)w(v)K_2(u,v)\,dv\,du$ in $Y[\rho]$, where $w(u) = \rho(u) - (1-\alpha)$. Riemann-sum convergence for the band sums, together with the integrability of the kernels, gives $Y_n/D_n \to F[\rho]$.

The feasibility conditions $|\rho(u)| < F[\rho]$ on each band ensure that the target moments are small enough for the associated Toeplitz moment problem to be solvable, so the Carathéodory discretization yields the required finite point set. Hence $C_{42} \le F[\rho] + \varepsilon$ for every admissible step-function $\rho$ and $\varepsilon > 0$, and the infimum statement follows. □

**Remark 2.** *Theorem 1 is the* upper-bound direction *of the continuum relaxation. It is exactly what we need for the numerical certificate: every certified value $F[\rho]$ gives a valid upper bound for $C_{42}$. The matching lower bound, which would show that the relaxation is sharp, is not required for the new bound and is left to future work.*

## 3 The $k$-band ansatz

We now restrict to step-function densities with at most $k$ active bands. Let $0 = t_0 < t_1 < \cdots < t_k < t_{k+1} = 1$ and assign a constant value $a_j \in \mathbb{C}$ to each band $[t_{j-1}, t_j]$. Writing $s = 1-\alpha$ on the first block $[0, t_1]$ and $w_j = a_j - s$, the functional $F[\rho] = |Y[\rho]|/D[\rho]$ becomes

$$Y = 1 - \sum_{j=1}^{k} w_j A_{1,j} + \frac{1}{2} \sum_{j,l=1}^{k} w_j w_l Q_{jl} + sK, \tag{3}$$

$$D = \int_0^{t_1} \frac{u^{\mathrm{Re}(\alpha)-1}}{1-u}\,du, \tag{4}$$

where

$$A_{1,j} = \int_{t_{j-1}}^{t_j} \frac{u^{\alpha-1}}{1-u}\,du, \qquad Q_{jl} = \int_{t_{j-1}}^{t_j} \int_{t_{l-1}}^{\min(t_l,\,1-u)} \frac{(1-u-v)^{\alpha-1}}{uv}\,dv\,du,$$

and $K = \int_0^{t_1} u^{\alpha-1}/(1-u)\,du$. A valid certificate additionally requires

$$|1-\alpha| < F[\rho] \quad \text{and} \quad |a_j| < F[\rho] \text{ for all } j \ge 1.$$

**Lemma 3** (Monotonicity)**.** *The subclass of $k$-band step functions is denoted $\mathcal{S}_k \subset \mathcal{P}$. Since $\mathcal{S}_k \subset \mathcal{S}_{k+1}$, the infimum $\inf_{\rho \in \mathcal{S}_k} F[\rho]$ is non-increasing in $k$.*

*Proof.* Immediate from the inclusion $\mathcal{S}_k \subset \mathcal{S}_{k+1}$ and the definition of the infimum. □

**Lemma 4** (Density)**.** *For every $\rho \in \mathcal{P}$ and every $\varepsilon > 0$, there exists a $k$-band step function $\rho_k \in \mathcal{S}_k$ with*

$$F[\rho_k] \le F[\rho] + \varepsilon.$$

*Proof sketch.* Restrict to a compact subclass $\mathcal{P}_M \subset \mathcal{P}$ defined by uniform bounds on $\tau^{-1}$, $\mathrm{Re}(\alpha)^{-1}$, $|\rho|$, and the uniform bound $M_\rho$. On $\mathcal{P}_M$, the functional $F$ is continuous with respect to uniform convergence away from $u = 0$ and the hypotenuse $u + v = 1$; the singularities are integrable, so small changes near the singular set contribute $O(\varepsilon)$. Since step functions are dense in $L^\infty$, any $\rho \in \mathcal{P}_M$ can be approximated by a $k$-band step function $\rho_k$, and the approximation can be refined until $F[\rho_k] \le F[\rho] + \varepsilon$. □

**Theorem 5** (Monotone convergence to the continuum optimum)**.** *The infimum over $k$-band step functions is monotone and converges to the continuum optimum:*

$$\inf_{\rho \in \mathcal{S}_k} F[\rho] \downarrow \inf_{\rho \in \mathcal{P}} F[\rho] \qquad \text{as } k \to \infty.$$

*Proof.* Monotonicity follows from Lemma 3. For the limit, Lemma 4 gives, for every $\rho \in \mathcal{P}$ and $\varepsilon > 0$, a $k$-band step function $\rho_k \in \mathcal{S}_k$ with $F[\rho_k] \le F[\rho] + \varepsilon$. Thus $\limsup_{k\to\infty} \inf_{\mathcal{S}_k} F \le \inf_{\mathcal{P}} F$, while the reverse inequality follows from $\mathcal{S}_k \subset \mathcal{P}$. □

Consequently, each certified $k$-band value is a valid upper bound for $C_{42}$ by Theorem 1, and the continuum optimum can be approached arbitrarily closely by taking $k$ large enough.

# 4 Interval computability

The final ingredient is that the k-band functional can be evaluated with rigorous interval arithmetic, so every certified bound is a genuine theorem rather than a numerical observation.

## Parameter box

For $k$ bands the parameter vector has $3k - 1$ real coordinates:

$$(t_1, f_2, \ldots, f_{k-1}, \mathrm{Re}\,\alpha, \mathrm{Im}\,\alpha, \mathrm{Re}\,\eta_2, \mathrm{Im}\,\eta_2, \ldots, \mathrm{Re}\,\eta_k, \mathrm{Im}\,\eta_k).$$

The internal cutoffs are recovered from the fractions by

$$t_j = t_1 + f_j(1 - 2t_1), \qquad 2 \le j \le k - 1,$$

and the symmetry condition fixes $t_k = 1 - t_1$. A certificate is computed on a box of half-width $h$ (typically $h = 10^{-12}$) centered at a candidate point. Every quantity below is evaluated with interval arithmetic, so the result is valid for the entire box.

## Interval arithmetic

All computations use the `mpmath.iv` interval library at 50 decimal digits of precision. Complex intervals are represented as `mpc`$(x, y)$, and every arithmetic operation is outward-rounded. This turns the functional evaluation into an interval enclosure $[C_{\mathrm{low}}, C_{\mathrm{high}}]$ valid for every parameter in the box.

### One-dimensional integrals

The 1D integrals $K$, $D$ and $A_{1,j}$ have the exact antiderivative

$$I_A(x) = \sum_{r \geq 0} \frac{x^{\alpha+r}}{\alpha+r}, \qquad \int_a^b \frac{u^{\alpha-1}}{1-u}\, du = I_A(b) - I_A(a).$$

In the interval implementation the infinite series is truncated after a finite number of terms and a rigorous geometric tail bound is added to the partial sum. Because $|x^{\alpha+r}| \leq x^{\mathrm{Re}(\alpha)+r}$ for $0 < x < 1$, the discarded tail is enclosed by an interval geometric series, giving a mathematically exact enclosure of $I_A(x)$.

### Two-dimensional integrals

The 2D integrals $Q_{jl}$ have an algebraic singularity along the hypotenuse $u+v=1$. After the graded substitution

$$v = V_{\max} - Ls^p, \qquad s \in [0,1],$$

with $V_{\max} = \min(t_l, 1-u)$ and $L = V_{\max} - t_{l-1}$, the transformed integrand is smooth. The integral is then evaluated with Gauss–Legendre quadrature in both variables; all quadrature weights include the correct outer Jacobian $(b-a)/2$ and the inner Jacobian $Lps^{p-1}$. Because every operation uses interval arithmetic, the result is a rigorous enclosure of the quadrature approximation.

### Validated remainder and composition

The Gauss–Legendre enclosures of Section 4 approximate the exact 2D integrals. A separate validated remainder bound, implemented in `c42_quadrature_remainder.py`, accounts for the truncation error of the product rule on each band pair. Summing the band-pair remainders gives an interval $[0, C_{\mathrm{rem}}]$.

For a parameter box the final rigorous certificate is

$$C_{42} \leq C_{\mathrm{high}} + C_{\mathrm{rem}},$$

where $C_{\mathrm{high}}$ is the interval upper bound on the quadrature-approximated functional and $C_{\mathrm{rem}}$ is the validated quadrature remainder. Both quantities are computed entirely with interval arithmetic, so the bound holds for every point in the box.

# 5 Explicit certificate

We instantiate the interval-computability framework of Section 4 with the certified $k=3$ point.

## 5.1 Candidate point and decoded density

Using differential evolution, Latin-hypercube sampling, and local Nelder–Mead refinement, we found a feasible $k=3$ point with parameter vector

$$(t_1,\, t_2,\, \alpha,\, \eta_2,\, \eta_3),$$

where the final cutoff is fixed by symmetry at $t_3 = 1 - t_1$. The numerical values are

$$\begin{aligned} t_1 &= 0.040511664590274124, \\ t_2 &= 0.959488204469002, \\ t_3 &= 0.9594883354097259, \\ \alpha &= 1.0047583326115217 + 0.363102621255547\, i, \\ \eta_2 &= 0.310410208083373 - 0.00036734153660538993\, i, \\ \eta_3 &= 0.22134893688471718 - 0.3323721384294311\, i. \end{aligned}$$

At this point the continuum functional evaluates to

$$C = 0.399333351092805, \qquad Y = 0.0097499082 - 0.0129227509\, i, \qquad D = 0.0405380911.$$

The active constraints are

$$|1-\alpha| = 0.3631337981, \quad |\eta_2| = 0.3104104254, \quad |\eta_3| = 0.3993326812,$$

so the constraint margin is $6.6 \times 10^{-7}$.

The structure is notable: the third band $[t_2, t_3]$ has width only $1.3 \times 10^{-4}$, producing a sharp boundary layer. Moreover $\alpha \approx 1 + 0.363i$, so $1 - \alpha$ is nearly pure imaginary.

### 5.2 Interval certificate

We certified the bound using the `mpmath.iv` interval arithmetic library at 50 decimal digits. The 1D integrals $K$, $D$ and $A_{1,j}$ are evaluated as exact antiderivatives via the series

$$I_A(x) = \sum_{r \geq 0} \frac{x^{\alpha+r}}{\alpha + r},$$

with a rigorous geometric tail bound added to the partial sum. The 2D integrals $Q_{jl}$ are carried out with Gauss–Legendre quadrature after the graded substitution $v = V_{\max} - Ls^p$, which removes the algebraic singularity at the hypotenuse $u + v = 1$. All quadrature weights include the correct $ba/2$ outer and $wj/2$ inner Jacobians.

For a parameter box of half-width $10^{-12}$ around the point above, the interval evaluation gives

$$C_{42} \leq C_{\text{iv}}^{\text{upper}} = 0.399333354781627.$$

The verdict is `CERTIFIED`: every point in the box satisfies the constraints and the interval part of the bound is below Griego's record.

### 5.3 Validated quadrature remainder

The certificate of the preceding subsection applies to the functional evaluated by Gauss–Legendre quadrature. To make the bound fully rigorous for the exact integrals, we add a validated remainder bound for the 2D quadrature error.

After the graded substitution, the transformed integrand is smooth on $[t_{j-1}, t_j] \times [0, 1]$. The product Gauss–Legendre error is bounded by an analytic-envelope estimate: if the transformed integrand is bounded by $M_0$ on a complex neighborhood of the integration domain and the nearest singularity is at distance $d$, then the $N$-node rule error is $O\big(M_0 L_u L_v (1+d/L)^{-2N}\big)$. Applying this to each band pair $(j,l)$ and summing gives a total quadrature remainder of

$$C_{\text{rem}} \leq 3.8663 \times 10^{-5}.$$

Adding this to the interval bound from Section 5.2 yields the fully rigorous certificate

$$\boxed{C_{42} \leq 0.399333354781627 + 3.8663 \times 10^{-5} = 0.3993720221096683}.$$

This improves Griego's record by 0.29128167 (approximately 42.2%).

# 6 Conclusion

We have proved that every admissible continuum density $\rho \in \mathcal{P}$ gives an asymptotic upper bound for Turán's pure power sum constant $C_{42}$, and we have certified an explicit three-band density producing

$$\boxed{C_{42} \leq 0.3993720221096683},$$

which breaks Griego's previous record of 0.690653695151631 by more than 42%. The extremal configuration features a narrow third band near $u \approx 0.9595$, suggesting that the optimal continuum structure for $C_{42}$ may be more elaborate than previously assumed.

# References


[1] Griego, R. J. *An upper bound for the Turán pure power sum problem.* Unpublished manuscript (cited in Erdős problem #519).

[2] Johansson, F. et al. *mpmath: a Python library for arbitrary-precision floating-point arithmetic.* Version 1.3.